

Bailey Forbes

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Software Engineer with security mindset — Enterprise Platform Development, Integration Architecture & Systems Programming

Full-stack engineer building scalable enterprise solutions spanning platform architecture, systems integration, and AI-assisted automation. Reduced compliance processes from **months to days**; architected multi-tenant platforms; and supported systems containing **584 million records**.

Professional Experience

Software Engineer Intern — Baker Hill — Integration Systems & Production Engineering

C#.NET, TypeScript, Azure DevOps, Salesforce | Indianapolis, IN **Sep 2025 – May 2026**
*Architect for enterprise integrations and supported production loan-origination systems serving **100+ financial institutions**.*

- Pioneered API pipeline reducing compliance turnaround from **months to days** by decoupling business logic from HTTP transport and automating data exchange via dependency injection and factory patterns
- Supported production systems with **584 million records** across **10+ institutions**: triaged escalations, validated CI/CD deployments, and approved hotfix merges
- Maintained integration reliability through Azure DevOps, Salesforce, and SRE best practices

Volunteer Platform Engineer & Lead Developer — [Homework Central Website Migration](#)

C#.NET, React, TypeScript, PostgreSQL, Rust, Docker | Sellersburg, IN **Jun 2026 – Sep 2026**
*Mission-driven platform architecture & AI integration: led migration of a **16,000-member Discord community** to an enterprise platform with RBAC and ML-assisted moderation, validated through a volunteer trial.*

- Architected C#.NET microservices, React/TypeScript interfaces, and PostgreSQL schemas for authentication, messaging, moderation, and RBAC
- Built AI-assisted moderation with **35,600+ trainable weights**, evaluation pipelines, and human-escalation workflows
- Engineered bit-packed permission masks for **constant-time authorization** checks
- Led instruction, oversight, and feedback collection for a **75-member** volunteer trial to evaluate usability and operations and decide whether the system was worth integrating

Selected Systems & Healthcare Projects

[Flintstone Kernel](#) — Creator and Lead Developer

2023 – Present

C, C++, Assembly, Python, Linux/UNIX, Hardware Abstraction, Networking

Portable kernel emphasizing backward compatibility and modular subsystem ownership: similar to enterprise platform version management and upgrade patterns.

- Engineered hardware abstraction, device drivers, and shell interfaces for physical and virtualized environments
- Maintained backward compatibility across **five releases** and **70+ contracts** with web-based QEMU boot testing
- Built custom TCP/IP and 802.11ax (Wi-Fi 6) stack for network communication including chat and file exchange

[KeyQuorum](#) — Creator and Lead Developer

2026 – Present

Rust, SQLite, Cryptography, RBAC, REST APIs, USB Security

Secure file-sharing system using hierarchical key quorums and RBAC, designed for USB-based hardware authentication.

- Engineered **2-of-3** hierarchical quorum access control with recursive key splitting, reconstruction, and revocation
- Built encrypted file sharing and co-signature workflows with role-based authorization and key rotation
- Designed a file history manager with custom envelope and file-tracking formats: owner- or ancestor-signed, strictly ordered, replay-protected updates
- Designed and tested secure client-relay architecture with encrypted messaging and SQLite persistence

Electronic Medical Record System — Healthcare Platform Engineer

Aug 2021 – May 2022

Kotlin, Java, REST APIs, Firebase, HIPAA-Focused Security Testing

*HIPAA-compliant healthcare platform spanning **26 hospital locations**: designed Kotlin patient workflows, Java staff services, and REST API contracts with cross-validated data flows.*

- Architected platform with Kotlin patient services, Java staff workflows, and REST APIs; cross-validated data consistency across local backups, Firebase, and REST endpoints
- Achieved **60% efficiency improvement** by redesigning workflows and validating healthcare data consistency
- Conducted HIPAA compliance analysis, penetration testing, and identified vulnerabilities; guided sponsor toward full administrative control

Education

Indiana University Southeast — B.S. Computer Science, Minor in Mathematics, Dean's List

May 2025

Indiana University Southeast — Cybersecurity Certificate, Information Security Track, Summa Cum Laude

Dec 2024

Technical Skills

Languages: C#, TypeScript, JavaScript, Python, SQL, Java, C, C++, Rust

Enterprise: .NET, REST APIs, Microservices, Integration Architecture, Design Patterns, Dependency Injection, RBAC

Tools: Azure DevOps, Docker, PostgreSQL, SQL Server, IIS, Git, CI/CD, SRE Practices